$$= \frac{(\sin A\cos A+1)\left(\frac{\sin A}{\sin A\cos A}-\frac{\cos A}{\sin A\cos A}\right)}{(\sec A-\operatorname{cosec}A)\left(\frac{\sin^2 A+\sin A\cos A+\cos^2 A}{\sin^2 A\cos^2 A}\right)}$$

$$= \frac{(\sin A\cos A+1)(\sec A-\cos ecA)}{(\sec A-\cos ecA)(1+\sin A\cos A)}\times \sin^2 A\cos^2 A = \sin^2 A\cos^2 A$$

Example 6.17 If $\frac{\cos^2\theta}{\sin\theta}=p$ and $\frac{\sin^2\theta}{\cos\theta}=q$, then prove that $p^2q^2(p^2+q^2+3)=1$

Solution We have $\frac{\cos^2\theta}{\sin\theta}=p$ ...(1) and $\frac{\sin^2\theta}{\cos\theta}=q$ ...(2)

$$p^2q^2(p^2+q^2+3) = \left(\frac{\cos^2\theta}{\sin\theta}\right)^2\left(\frac{\sin^2\theta}{\cos\theta}\right)^2\times\left[\left(\frac{\cos^2\theta}{\sin\theta}\right)^2+\left(\frac{\sin^2\theta}{\cos\theta}\right)^2+3\right] \text{ [from (1) and (2)]}$$

$$= \left(\frac{\cos^4\theta}{\sin^2\theta}\right)\left(\frac{\sin^4\theta}{\cos^2\theta}\right)\times\left[\frac{\cos^4\theta}{\sin^2\theta}+\frac{\sin^4\theta}{\cos^2\theta}+3\right]$$

$$= (\cos^2\theta\times\sin^2\theta)\times\left[\left(\frac{\cos^6\theta+\sin^6\theta+3\sin^2\theta\cos^2\theta}{\sin^2\theta\cos^2\theta}\right)\right]$$

$$= \cos^6\theta+\sin^6\theta+3\sin^2\theta\cos^2\theta$$

$$= (\cos^2\theta)^3+(\sin^2\theta)^3+3\sin^2\theta\cos^2\theta$$

$$= [(\cos^2\theta+\sin^2\theta)^3-3\cos^2\theta\sin^2\theta(\cos^2\theta+\sin^2\theta)]+3\sin^2\theta\cos^2\theta$$

$$= 1-3\cos^2\theta\sin^2\theta(1)+3\cos^2\theta\sin^2\theta = 1$$

Progress Check

1. The number of trigonometric ratios is _____.
2. $1-\cos^2\theta$ is _____.
3. $(\sec\theta+\tan\theta)(\sec\theta-\tan\theta)$ is _____.
4. $(\cot\theta+\operatorname{cosec}\theta)(\cot\theta-\operatorname{cosec}\theta)$ is _____.
5. $\cos 60^\circ\sin 30^\circ+\cos 30^\circ\sin 60^\circ$ is _____.
6. $\tan 60^\circ\cos 60^\circ+\cot 60^\circ\sin 60^\circ$ is _____.
7. $(\tan 45^\circ+\cot 45^\circ)+(\sec 45^\circ\operatorname{cosec}45^\circ)$ is _____.
8. (i) $\sec\theta=\operatorname{cosec}\theta$ if θ is _____. (ii) $\cot\theta=\tan\theta$ if θ is _____.

Exercise 6.1

1. Prove the following identities.
   (i) $\cot\theta+\tan\theta=\sec\theta\operatorname{cosec}\theta$ (ii) $\tan^4\theta+\tan^2\theta=\sec^4\theta-\sec^2\theta$
2. Prove the following identities.
   (i) $\frac{1-\tan^2\theta}{\cot^2\theta-1}=\tan^2\theta$ (ii) $\frac{\cos\theta}{1+\sin\theta}=\sec\theta-\tan\theta$

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